Physical Chemistry (I)

Lecture 12

Classical Statistical Mechanics

In the Last Class

- Maxwell-Boltzmann statistics
  - Boltzmann factor
  - Partition function
- Average quantities
  - Average energy
  - Average entropy: Shannon entropy
  - Partition function and thermodynamics
- Phase space

Last class

Maxwell-Boltzmann Statistics

Average distribution of independent particles

of a system in thermal equilibrium

Described by the Boltzmann factor

$$P(s) \propto e^{-E(s)/k_{\mathrm{B}}T}$$

Last class

Partition Function

$$P(s) = \frac{1}{z} e^{-E(s)/k_{\mathrm{B}}T}$$

Using the normalization condition, $\sum_s P(s) = 1$,

$$1 = \sum_s \frac{1}{z} e^{-E(s)/k_{\mathrm{B}}T} = \frac{1}{z} \sum_s e^{-E(s)/k_{\mathrm{B}}T}$$

$$z = \sum_s e^{-E(s)/k_{\mathrm{B}}T}$$

(Partition function)

Average Energy

$$\langle E \rangle = \sum_i P(E_i)E_i = \sum_i \frac{E_i e^{-\beta E_i}}{z}$$

$$\frac{\partial \ln z}{\partial \beta} = \frac{1}{z}\frac{\partial z}{\partial \beta} = -\frac{1}{z}\sum_i E_i e^{-\beta E_i}$$

Hence,

$$\langle E \rangle = U = -\frac{\partial \ln z}{\partial \beta}$$

Average Entropy

$S = -k_\mathrm{B} \sum_i p_i \ln p_i$ (Shannon entropy)

Substitution of $p_i = P(E_i) = \frac{1}{z} e^{-\beta E_i}$ leads to

$$S = \frac{\langle E \rangle}{T} + k_\mathrm{B} \ln z$$

Therefore,

$$-k_\mathrm{B} T \ln z = \langle E \rangle - TS = A$$

Partition Function and …

- Helmholtz energy: $A = -k_\mathrm{B}T \ln z$
- Internal energy: $U = -\partial(\ln z)/\partial\beta$
- Entropy: $S = (U - A)/T$

"A partition function contains

all the thermodynamic information about a system."

Last class

Derived Functions

We know Helmholtz energy from the partition function,

$$A = -k_{\mathrm{B}} T \ln z$$

Using $\mathrm{d}A = -S\mathrm{d}T - p\mathrm{d}V$,

$$S = -\left(\frac{\partial A}{\partial T}\right)_V, \qquad p = -\left(\frac{\partial A}{\partial V}\right)_T$$

Example: Spin System

Assume that we have a single spin system: up (↑) or down (↓)

$$E(\uparrow) = +\frac{\epsilon}{2}, \qquad E(\downarrow) = -\frac{\epsilon}{2}$$

The partition function is

$$z = \sum_s e^{-E(s)/k_\mathrm{B}T} = e^{-E(\uparrow)/k_\mathrm{B}T} + e^{-E(\downarrow)/k_\mathrm{B}T}$$

$$z = e^{-\epsilon/2k_\mathrm{B}T} + e^{+\epsilon/2k_\mathrm{B}T}$$

Phase Space

Space of all possible microstates of a system

- Classical mechanics:

Space of all possible values of

position and momentum variables

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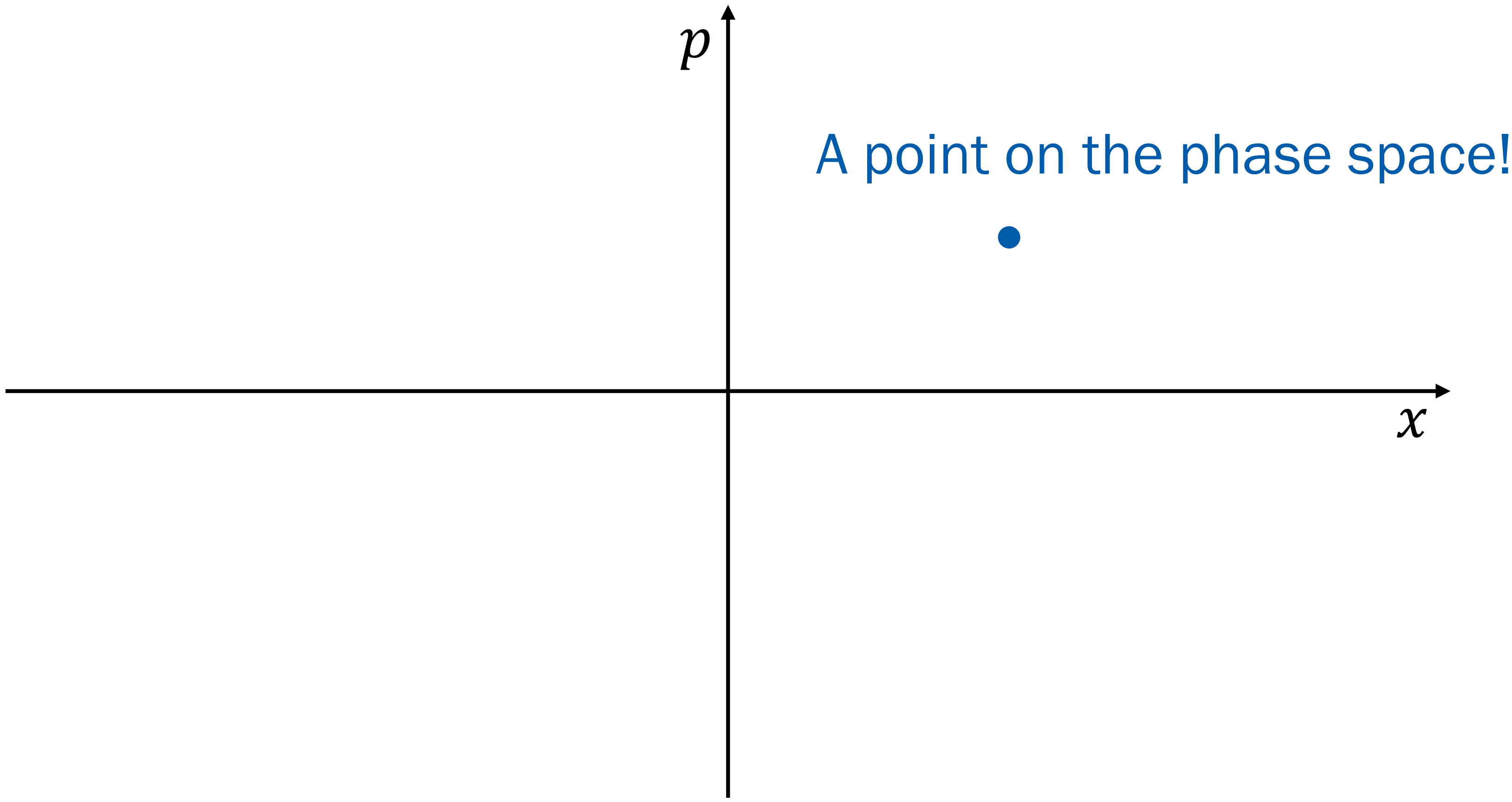
Microstate of Classical Particle



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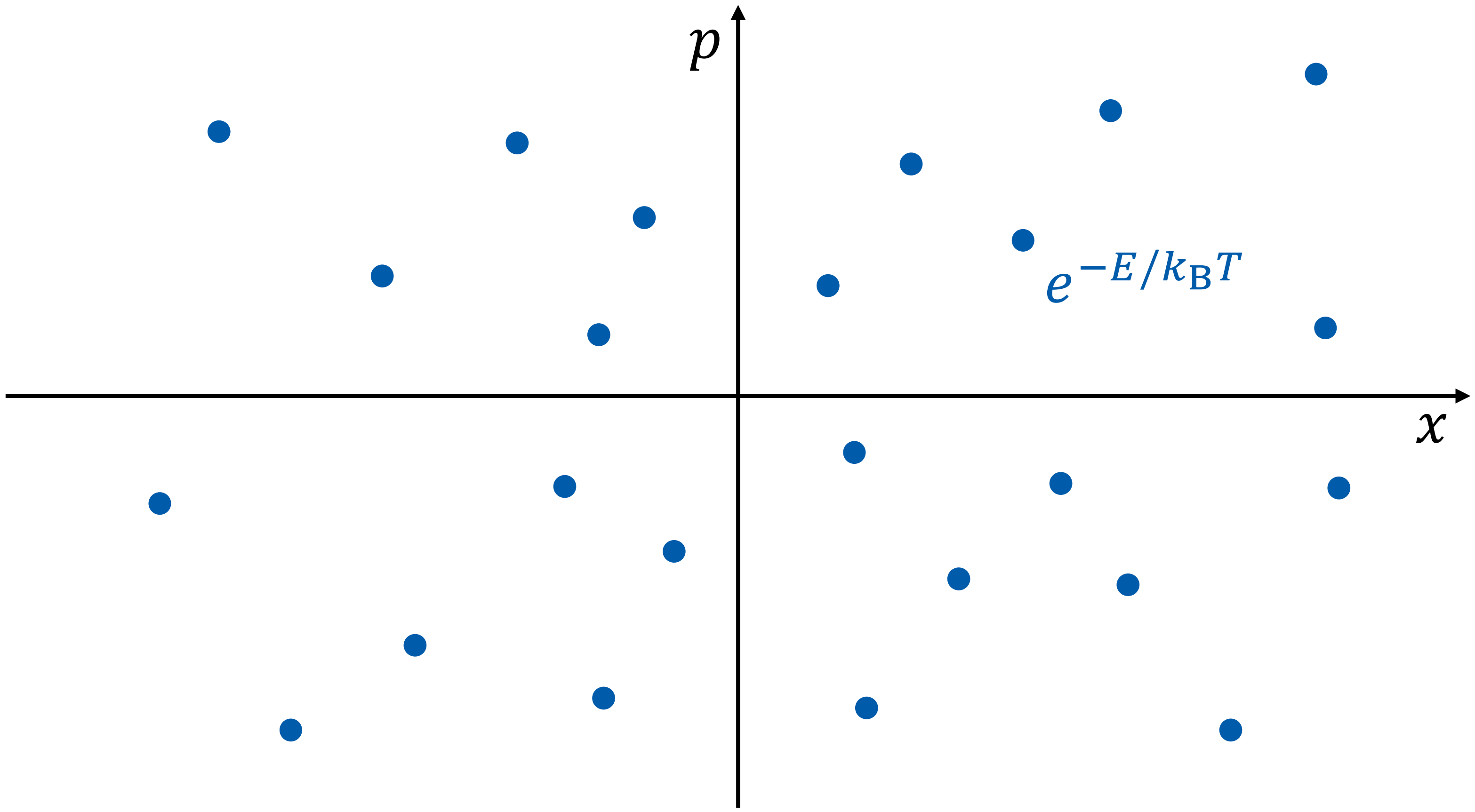
Partition Function



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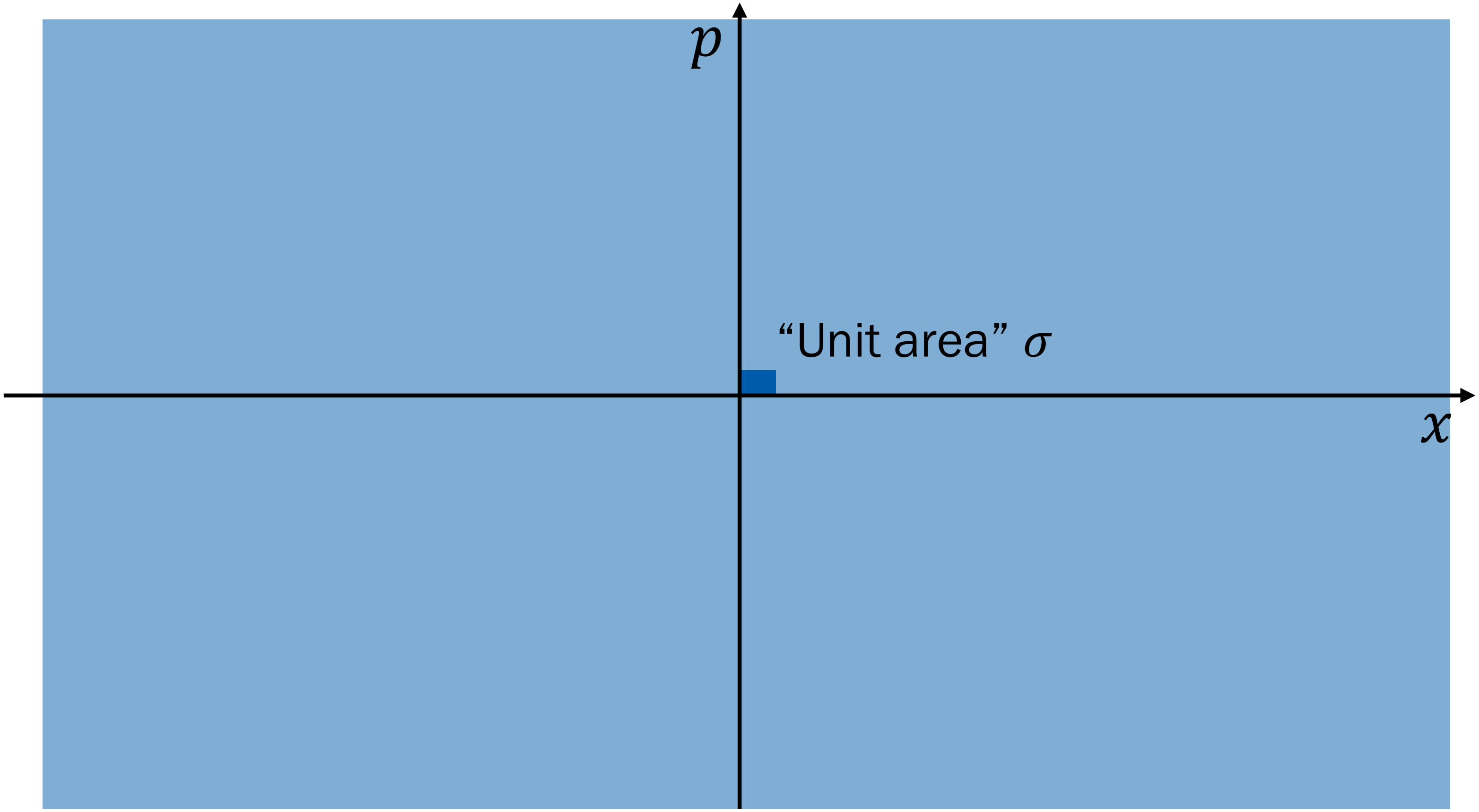
Partition Function



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Partition Function

$$z = \sum_{s} e^{-E(s)/k_{\mathrm{B}}T} \;\rightarrow\; z = \frac{1}{\sigma}\int\int e^{-E(x,p)/k_{\mathrm{B}}T}\,\mathrm{d}x\mathrm{d}p$$

for a 1-dimensional single particle

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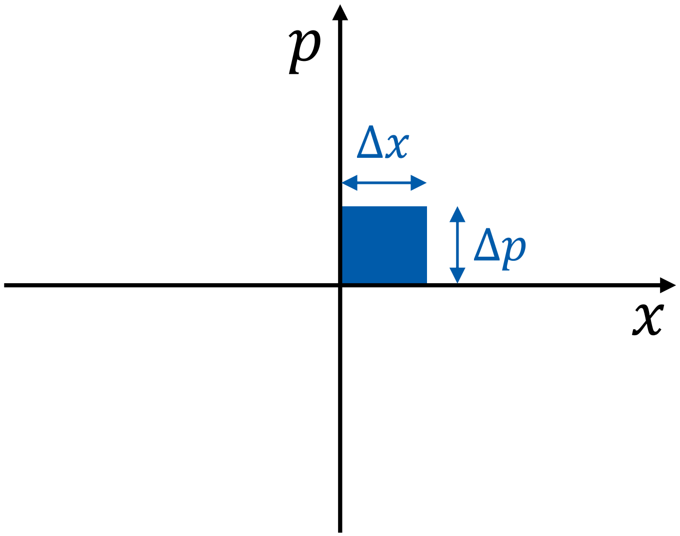
What is σ?

From Heisenberg's uncertainty principle,

$\Delta x \Delta p \geq h$ (Planck's constant)



Take $\sigma = h$!

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Partition Function

$$z = \frac{1}{h} \int \int e^{-E(x,p)/k_{\mathrm{B}}T} \, \mathrm{d}x \mathrm{d}p$$

- Example: 1-dimensional spring

$$E(x,p) = \frac{1}{2}kx^2 + \frac{1}{2}mv^2 = \frac{1}{2}kx^2 + \frac{p^2}{2m}$$

$$z = \frac{1}{h} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} e^{-\frac{1}{k_{\mathrm{B}}T}\left(\frac{1}{2}kx^2 + \frac{p^2}{2m}\right)} \, \mathrm{d}x \mathrm{d}p$$

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1-Dimensional Spring

$$z = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{1}{k_{\mathrm{B}}T}\left(\frac{1}{2}kx^2+\frac{p^2}{2m}\right)}\,\mathrm{d}x\mathrm{d}p$$

$$z = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_{\mathrm{B}}T}-\frac{p^2}{2mk_{\mathrm{B}}T}}\,\mathrm{d}x\mathrm{d}p$$

$$z = \frac{1}{h}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_{\mathrm{B}}T}} \times e^{-\frac{p^2}{2mk_{\mathrm{B}}T}}\,\mathrm{d}x\mathrm{d}p$$

$$z = \frac{1}{h}\left[\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_{\mathrm{B}}T}}\,\mathrm{d}x\right]\left[\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mk_{\mathrm{B}}T}}\,\mathrm{d}p\right]$$

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Gaussian Integrals

$$z = \frac{1}{h}\left[\int_{-\infty}^{\infty} e^{-\frac{kx^2}{2k_\mathrm{B}T}}\,\mathrm{d}x\right]\left[\int_{-\infty}^{\infty} e^{-\frac{p^2}{2mk_\mathrm{B}T}}\,\mathrm{d}p\right]$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2}\,\mathrm{d}x = \sqrt{\pi/\alpha}$,

$$z = \frac{1}{h}\sqrt{\frac{2\pi k_\mathrm{B}T}{k}}\sqrt{2m\pi k_\mathrm{B}T} = \frac{2\pi k_\mathrm{B}T}{h}\sqrt{\frac{m}{k}}$$

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Into Higher Dimensions

- Single 1-dimensional particle

$$z = \frac{1}{h}\int\int e^{-E(x,p)/k_\mathrm{B}T}\,\mathrm{d}x\mathrm{d}p$$

- Single 3-dimensional particle

$$z = \frac{1}{h^3}\iiint\iiint e^{-E(x,y,z,p_x,p_y,p_z)/k_\mathrm{B}T}\,\mathrm{d}x\mathrm{d}y\mathrm{d}z\mathrm{d}p_x\mathrm{d}p_y\mathrm{d}p_z$$

$$\rightarrow\ z = \frac{1}{h^3}\int\int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\,\mathrm{d}\vec{r}\mathrm{d}\vec{p}$$

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Higher and Higher

- Single 3-dimensional particle

$$z = \frac{1}{h^3} \int \int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T} \,\mathrm{d}\vec{r}\mathrm{d}\vec{p}$$

- N 3-dimensional (independent) particles

$$Z = z_1 \times z_2 \times z_3 \times z_4 \times \cdots \times z_N$$

$$Z = \frac{1}{h^{3N}} \left[\int \int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T} \,\mathrm{d}\vec{r}\mathrm{d}\vec{p} \right]^N$$

Indistinguishability

- N 3-dimensional distinguishable particles:

$$Z = \frac{1}{h^{3N}}\left[\int\int e^{-E(\vec{r},\vec{p})/k_{\mathrm{B}}T}\,\mathrm{d}\vec{r}\mathrm{d}\vec{p}\right]^N$$

- N 3-dimensional indistinguishable particles:

$$Z = \frac{1}{N!\,h^{3N}}\left[\int\int e^{-E(\vec{r},\vec{p})/k_{\mathrm{B}}T}\,\mathrm{d}\vec{r}\mathrm{d}\vec{p}\right]^N$$

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Calculation of Z

For indistinguishable particles,

$$Z = \frac{1}{N!\,h^{3N}}\left[\int\int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\,\mathrm{d}\vec{r}\mathrm{d}\vec{p}\right]^N = \frac{1}{N!}z^N$$

where

$$z = \frac{1}{h^3}\int\int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\,\mathrm{d}\vec{r}\mathrm{d}\vec{p}$$

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Perfect Gas

- No interactions (no potential energy)
- Total energy = kinetic energy ($p^2/2m$)
- Bound in a box of volume V
  - Range of x = $(0, a)$
  - Range of y = $(0, b)$
  - Range of z = $(0, c)$
  - $abc = V$
- Range of momentum = $(-\infty, \infty)$

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z of Perfect Gas

$$z = \frac{1}{h^3} \int \int e^{-E(\vec{r},\vec{p})/k_\mathrm{B}T}\, \mathrm{d}\vec{r}\mathrm{d}\vec{p}$$

Since $E(\vec{r},\vec{p}) = p^2/2m$,

$$z = \frac{1}{h^3} \int \int e^{-p^2/2mk_\mathrm{B}T}\, \mathrm{d}\vec{r}\mathrm{d}\vec{p}$$

$$z = \frac{1}{h^3} \left[\int 1\, \mathrm{d}\vec{r} \right] \left[\int e^{-p^2/2mk_\mathrm{B}T}\, \mathrm{d}\vec{p} \right]$$

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Position Integral

$$z = \frac{1}{h^3}\left[\int 1\,\mathrm{d}\vec{r}\right]\left[\int e^{-p^2/2mk_\mathrm{B}T}\,\mathrm{d}\vec{p}\right]$$

$$\int 1\,\mathrm{d}\vec{r} = \int_0^c \int_0^b \int_0^a 1\,\mathrm{d}x\,\mathrm{d}y\,\mathrm{d}z = abc = V$$

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Momentum Integral

$$z = \frac{V}{h^3}\left[\int e^{-p^2/2mk_\mathrm{B}T}\,\mathrm{d}\vec{p}\right]$$

$$\int e^{-p^2/2mk_\mathrm{B}T}\,\mathrm{d}\vec{p} = \int_{-\infty}^{\infty}\int_{-\infty}^{\infty}\int_{-\infty}^{\infty} e^{-p^2/2mk_\mathrm{B}T}\,\mathrm{d}p_x\,\mathrm{d}p_y\,\mathrm{d}p_z$$

Since $e^{-p^2/2mk_\mathrm{B}T} = e^{-p_x^2/2mk_\mathrm{B}T} e^{-p_y^2/2mk_\mathrm{B}T} e^{-p_z^2/m2k_\mathrm{B}T}$,

$$\left[\int_{-\infty}^{\infty} e^{-p_x^2/2mk_\mathrm{B}T}\,\mathrm{d}p_x\right]\left[\int_{-\infty}^{\infty} e^{-p_y^2/2mk_\mathrm{B}T}\,\mathrm{d}p_y\right]\left[\int_{-\infty}^{\infty} e^{-p_z^2/2mk_\mathrm{B}T}\,\mathrm{d}p_z\right]$$

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Gaussian Integral Again

$$z = \frac{V}{h^3}\left[\int_{-\infty}^{\infty} e^{-p_x^2/2mk_\mathrm{B}T}\,\mathrm{d}p_x\right]^3$$

Using $\int_{-\infty}^{\infty} e^{-\alpha x^2}\,\mathrm{d}x = \sqrt{\pi/\alpha}$,

$$z = \frac{V}{h^3}\left[\sqrt{2\pi m k_\mathrm{B}T}\right]^3 = V\left(\frac{2\pi m k_\mathrm{B}T}{h^2}\right)^{3/2}$$

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Z of Perfect Gas

$$z = V\left(\frac{2\pi m k_{\mathrm{B}} T}{h^2}\right)^{3/2}$$

For indistinguishable gas particles,

$$Z = \frac{1}{N!} z^N = \frac{V^N}{N!}\left(\frac{2\pi m k_{\mathrm{B}} T}{h^2}\right)^{3N/2}$$

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Helmholtz Energy

$$Z = \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)^{3N/2}$$

Now, we can calculate the Helmholtz energy:

$$A = -k_\mathrm{B} T \ln Z = -k_\mathrm{B} T \ln \left\{\frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)^{3N/2}\right\}$$

$$A = -N k_\mathrm{B} T \ln V + k_\mathrm{B} T \ln N! - \frac{3}{2} N k_\mathrm{B} T \ln \left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)$$

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Internal Energy

$$Z = \frac{V^N}{N!}\left(\frac{2\pi m k_\mathrm{B} T}{h^2}\right)^{3N/2}$$

Also the internal energy:

$$U = -\frac{\partial(\ln Z)}{\partial \beta} = -\frac{\partial}{\partial \beta}\ln\left\{\frac{V^N}{N!}\left(\frac{2\pi m}{\beta h^2}\right)^{3N/2}\right\}$$

$$U = \frac{\partial}{\partial \beta}\ln \beta^{3N/2} + \frac{\partial}{\partial \beta}(\text{terms independent to } \beta) = \frac{3}{2}N k_\mathrm{B} T$$

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Pressure

$$A = -Nk_{\mathrm{B}}T\ln V + k_{\mathrm{B}}T\ln N! - \frac{3}{2}Nk_{\mathrm{B}}T\ln\left(\frac{2\pi m k_{\mathrm{B}}T}{h^2}\right)$$

Using $p = -(\partial A/\partial V)_T$,

$$p = Nk_{\mathrm{B}}T\frac{\partial(\ln V)}{\partial V} + \frac{\partial}{\partial V}(\text{terms independent to } V)$$

$$p = \frac{Nk_{\mathrm{B}}T}{V}$$

One can obtain entropy similarly

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Real Gas

Particles are no more independent to each other

$$Z \neq z^N / N!$$

$$Z = \frac{1}{N!\, h^{3N}} \int \cdots \int e^{-E_{\mathrm{tot}}/k_{\mathrm{B}}T}\, \mathrm{d}\vec{r}_1 \mathrm{d}\vec{p}_1 \mathrm{d}\vec{r}_2 \mathrm{d}\vec{p}_2 \cdots \mathrm{d}\vec{r}_N \mathrm{d}\vec{p}_N$$

where

$$E_{\mathrm{tot}} = \sum_i \frac{p_i^2}{2m} + U(\vec{r}_1, \vec{r}_2, \cdots, \vec{r}_N)$$

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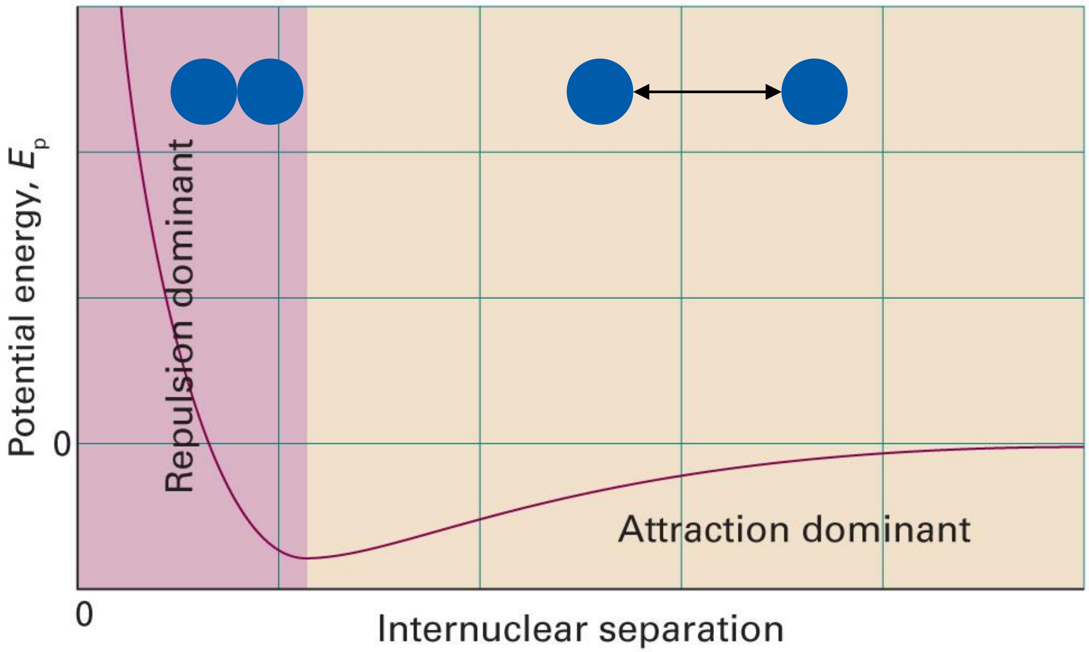
Potential Energy

Pairwise interactions (attraction + repulsion)



(Image: Atkins *et al.*, ***Atkins' Physical Chemistry***, 12th ed., Figure 1C.1)

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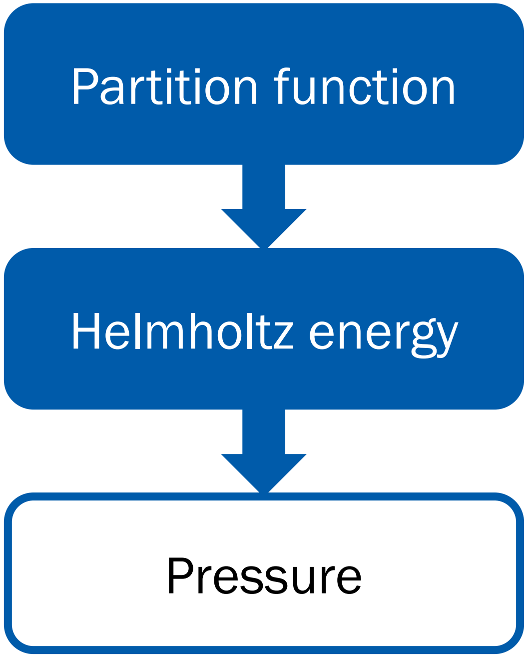
Pressure of Real Gas



$$p = \frac{nRT}{V - nb} - a\frac{n^2}{V^2}$$

Equipartition Theorem

For a sample at thermal equilibrium,

the average value of each quadratic contribution

to the internal energy is $(1/2)k_\mathrm{B}T$.

(Quadratic contribution: e.g., $E \propto p_x^2$ or $E \propto x^2$)

Can we prove this theorem?

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Setting up the Stage

$$Z = \frac{1}{N!\,h^{3N}} \int \cdots \int e^{-E_{\text{tot}}/k_{\text{B}}T}\, \mathrm{d}\vec{r}_1 \mathrm{d}\vec{p}_1 \mathrm{d}\vec{r}_2 \mathrm{d}\vec{p}_2 \cdots \mathrm{d}\vec{r}_N \mathrm{d}\vec{p}_N$$

Assume that

$$E_{\text{tot}} = \sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2$$

$$(A_i, B_j > 0)$$

and that the ranges of both positions and momenta are $(-\infty, \infty)$

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Summation to Product

$$Z = \frac{1}{N!\,h^{3N}} \int \cdots \int e^{-\left(\sum_{i=1}^{f_r} A_i r_i^2 + \sum_{j=1}^{f_p} B_j p_j^2\right)/k_\mathrm{B}T}\, \mathrm{d}\vec{r}_1 \mathrm{d}\vec{p}_1 \mathrm{d}\vec{r}_2 \mathrm{d}\vec{p}_2 \cdots \mathrm{d}\vec{r}_N \mathrm{d}\vec{p}_N$$

$$= \frac{1}{N!\,h^{3N}} \int \cdots \int \prod_{i=1}^{f_r} e^{-A_i r_i^2/k_\mathrm{B}T} \prod_{j=1}^{f_p} e^{-B_j p_j^2/k_\mathrm{B}T}\, \mathrm{d}\vec{r}_1 \mathrm{d}\vec{p}_1 \mathrm{d}\vec{r}_2 \mathrm{d}\vec{p}_2 \cdots \mathrm{d}\vec{r}_N \mathrm{d}\vec{p}_N$$

Thus,

$$Z = \frac{1}{N!\,h^{3N}} \prod_{i=1}^{f_r} \int_{-\infty}^{\infty} e^{-A_i r_i^2/k_\mathrm{B}T}\, \mathrm{d}r_i \prod_{j=1}^{f_p} \int_{-\infty}^{\infty} e^{-B_j p_j^2/k_\mathrm{B}T}\, \mathrm{d}p_j$$

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Gaussian Integrals

$$Z = \frac{1}{N!\,h^{3N}} \prod_{i=1}^{f_r} \int_{-\infty}^{\infty} e^{-A_i r_i^2/k_\mathrm{B}T}\,\mathrm{d}r_i \prod_{j=1}^{f_p} \int_{-\infty}^{\infty} e^{-B_j p_j^2/k_\mathrm{B}T}\,\mathrm{d}p_j$$

Using

$$\int_{-\infty}^{\infty} e^{-A_i r_i^2/k_\mathrm{B}T}\,\mathrm{d}r_i = \sqrt{\frac{\pi k_\mathrm{B}T}{A_i}}, \qquad \int_{-\infty}^{\infty} e^{-B_j p_j^2/k_\mathrm{B}T}\,\mathrm{d}p_j = \sqrt{\frac{\pi k_\mathrm{B}T}{B_j}},$$

we obtain

$$Z = \frac{1}{N!\,h^{3N}} \prod_{i=1}^{f_r} \left(\frac{\pi k_\mathrm{B}T}{A_i}\right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{\pi k_\mathrm{B}T}{B_j}\right)^{1/2}$$

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In Terms of $\boldsymbol{\beta}$

$$Z = \frac{1}{N!\,h^{3N}} \prod_{i=1}^{f_r} \left(\frac{\pi k_\mathrm{B} T}{A_i}\right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{\pi k_\mathrm{B} T}{B_j}\right)^{1/2}$$

We can write the partition function as follows:

$$Z = \frac{\pi^{(f_r+f_p)/2}}{N!\,h^{3N}} \left(\frac{1}{\beta}\right)^{(f_r+f_p)/2} \prod_{i=1}^{f_r} \left(\frac{1}{A_i}\right)^{1/2} \prod_{j=1}^{f_p} \left(\frac{1}{B_j}\right)^{1/2}$$

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Internal Energy

$$Z = \frac{\pi^{(f_r+f_p)/2}}{N!\,h^{3N}}\left(\frac{1}{\beta}\right)^{(f_r+f_p)/2}\prod_{i=1}^{f_r}\left(\frac{1}{A_i}\right)^{1/2}\prod_{j=1}^{f_p}\left(\frac{1}{B_j}\right)^{1/2}$$

The internal energy can be calculated from

$$U = -\left(\frac{\partial \ln Z}{\partial \beta}\right)$$

Now,

$$\ln Z = -\frac{f_r + f_p}{2}\ln\beta + (\text{terms independent to } \beta)$$

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Internal Energy

$$\ln Z = -\frac{f_r + f_p}{2} \ln \beta + (\text{terms independent to } \beta)$$

$$U = -\left(\frac{\partial \ln Z}{\partial \beta}\right) = \frac{f_r + f_p}{2} \frac{\partial \ln \beta}{\partial \beta} = \frac{f_r + f_p}{2} \frac{1}{\beta} = \frac{f_r + f_p}{2} k_\mathrm{B} T$$

Therefore,

$$U = \left(f_r + f_p\right) \frac{k_\mathrm{B} T}{2}$$

"The average value of each quadratic contribution

to the internal energy is $(1/2)k_\mathrm{B}T$."

Summary

$$S = k_{\mathrm{B}} \ln W$$

Boltzmann factor

$$P(s) = \frac{1}{z} e^{-E(s)/k_{\mathrm{B}}T}$$

- Partition function

$$z = \sum e^{-E(s)/k_{\mathrm{B}}T}$$

Classical Stat Mech

Phase space

Partition function

$$z = \frac{1}{h^3} \iint e^{-E/k_{\mathrm{B}}T} \, \mathrm{d}\vec{r}\mathrm{d}\vec{p}$$

- Many-particle PF

$$Z = \frac{1}{N!} z^N$$

Perfect gas

- Partition function
- Helmholtz energy
- Pressure

Perfect gas law

Real gas

Equipartition theorem

- Partition function
- Internal energy

Thermodynamics

Helmholtz energy, internal energy, …